

§3.4

Boxplots and Five- Number Summary

Five numbers, and the picture they draw

Today's Goal

- Calculate the five-number summary for a dataset
- Construct a boxplot from the five-number summary
- Interpret boxplots to understand data distribution
- Use the interquartile range to identify potential outliers
- Compare distributions using side-by-side boxplots

What You Will Be Graded On

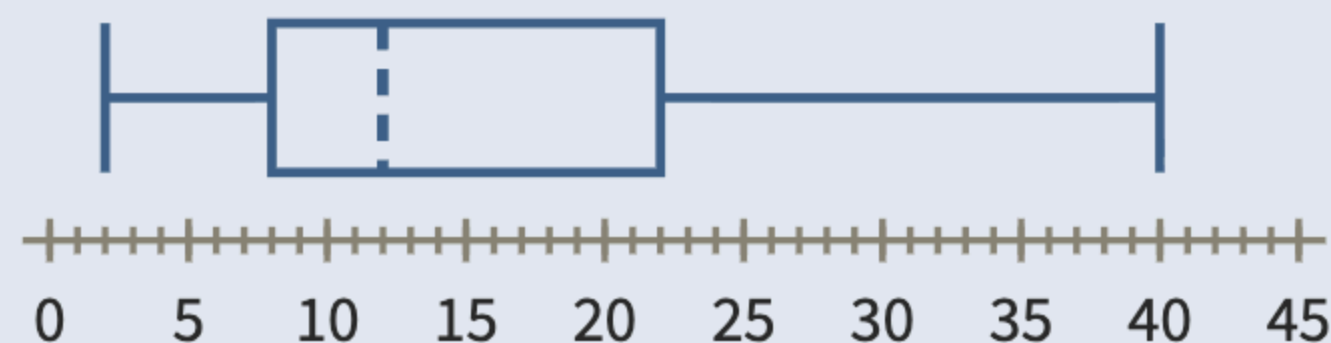
LT.5 Graphical Summaries of Data

I understand the different types of graphical displays for data (e.g., bar graphs, histograms, boxplots, etc.). I can interpret and create appropriate graphical summaries for various types of data.

LT.6 Numerical Summaries of Data

I understand the different numerical summaries for data (e.g., mean, median, mode, standard deviation, etc.). I can find and interpret the values of these numerical summaries for a given data set.

Minutes to Find a Parking Spot



**What can you tell me
about these times?**

The box holds the middle half of the times, and the line inside it is the median.

The whiskers reach out to the shortest and the longest time.

Cutting the Data Into Four

Definition 3.4.1

A *quartile* is one of three numbers used to divide a set of quantitative data into four groups with about 25% of the values in each group.

- Q_1 (First Quartile). This is approximately equivalent to the 25-th percentile.
- Q_2 (Second Quartile). This is approximately equivalent to the 50-th percentile, which is also the median.
- Q_3 (Third Quartile). This is approximately equivalent to the 75-th percentile.

The Five Numbers

Together with the minimum and maximum values, these quartile markers make up what we refer to as the *5-Number Summary* for a dataset.

Definition 3.4.2

The *five-number summary* of a quantitative data set consists of the minimum value, the three quartiles, and the maximum value:

$$\text{Min}, \quad Q_1, \quad Q_2, \quad Q_3, \quad \text{Max}.$$

Together these five numbers summarize both the center and the spread of the data and form the basis of a boxplot.

Finding the Five Numbers

Procedure 3.4.3 Computing the Five-Number Summary

1. Sort the data from smallest to largest.
2. Identify the minimum (first value) and maximum (last value).
3. Find the median Q_2 .

Then Split at the Median

Procedure 3.4.3, continued

4. Split the sorted data into a lower half (values below the median) and an upper half (values above the median). When n is odd, exclude the median from both halves.
 5. Find the median of the lower half to get Q_1 .
 6. Find the median of the upper half to get Q_3 .
- The five-number summary is Min, Q_1 , Q_2 , Q_3 , Max.

Five Numbers From Sorted Data

A campus bike shop logged the number of repairs finished on each of 15 weekdays.

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value	12	14	15	17	18	18	20	21	23	24	26	27	29	31	35

Find the five-number summary.

Step 1 is already done for you. The data is sorted.

Splitting at the Median

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Value	12	14	15	17	18	18	20	21	23	24	26	27	29	31	35

The first and last values give $\text{Min} = 12$ and $\text{Max} = 35$.

There are 15 values, so the median is the 8th value: $Q_2 = 21$.

Lower half, positions 1 through 7. Its median is the 4th value: $Q_1 = 17$.

Upper half, positions 9 through 15. Its median is the 4th of those: $Q_3 = 27$.

Five-number summary: 12, 17, 21, 27, 35.

How Wide Is the Middle Half?

Definition 3.4.5

The *interquartile range* of a quantitative dataset is defined as

$$\text{IQR} = Q_3 - Q_1$$

For the bike shop: $\text{IQR} = 27 - 17 = 10$ repairs.

The range was $35 - 12 = 23$. One unusual day can move the range. It cannot move the IQR.

When Is a Value Too Far Out?

The interquartile range provides us with a way to identify possible outliers. A value is considered to be a possible outlier if it is more than $1.5 \cdot \text{IQR}$ below Q_1 or more than $1.5 \cdot \text{IQR}$ above Q_3 .

Bike shop: $\text{IQR} = 10$, so $1.5 \cdot \text{IQR} = 15$.

Thresholds $17 - 15 = 2$ and $27 + 15 = 42$. Every value sits between them, so nothing is flagged.

The 1.5 Times IQR Rule

Procedure 3.4.6 Identifying Possible Outliers Using the 1.5×IQR Rule

1. Compute Q_1 and Q_3 (or use the five-number summary).
2. Compute the IQR: $IQR = Q_3 - Q_1$.
3. Compute the lower threshold: $Q_1 - 1.5 \cdot IQR$.
4. Compute the upper threshold: $Q_3 + 1.5 \cdot IQR$.
5. Flag as a possible outlier any data value that lies *below* the lower threshold or *above* the upper threshold.

One Very Busy Shift

A food truck recorded the number of orders served in each of 11 lunch shifts.

Position	1	2	3	4	5	6	7	8	9	10	11
Value	34	36	38	39	41	42	44	45	47	48	96

Find the five-number summary, then check for possible outliers.

Running the 1.5 Times IQR Rule

Position	1	2	3	4	5	6	7	8	9	10	11
Value	34	36	38	39	41	42	44	45	47	48	96

There are 11 values, so the median is the 6th. Summary: 34, 38, 42, 47, 96.

$\text{IQR} = 47 - 38 = 9$, so $1.5 \cdot \text{IQR} = 13.5$.

Lower threshold: $38 - 13.5 = 24.5$. Upper threshold: $47 + 13.5 = 60.5$.

Nothing sits below 24.5. The value 96 sits above 60.5, so it is flagged.

Flagged means look into it. It does not mean throw it away.

Drawing the Five Numbers

Definition 3.4.7

A *boxplot* (also called a *box-and-whisker plot*) is a graphical display of data consisting of a line extending from the minimum value to the maximum value, with a box drawn from Q_1 to Q_3 and a line inside the box marking the median (Q_2).

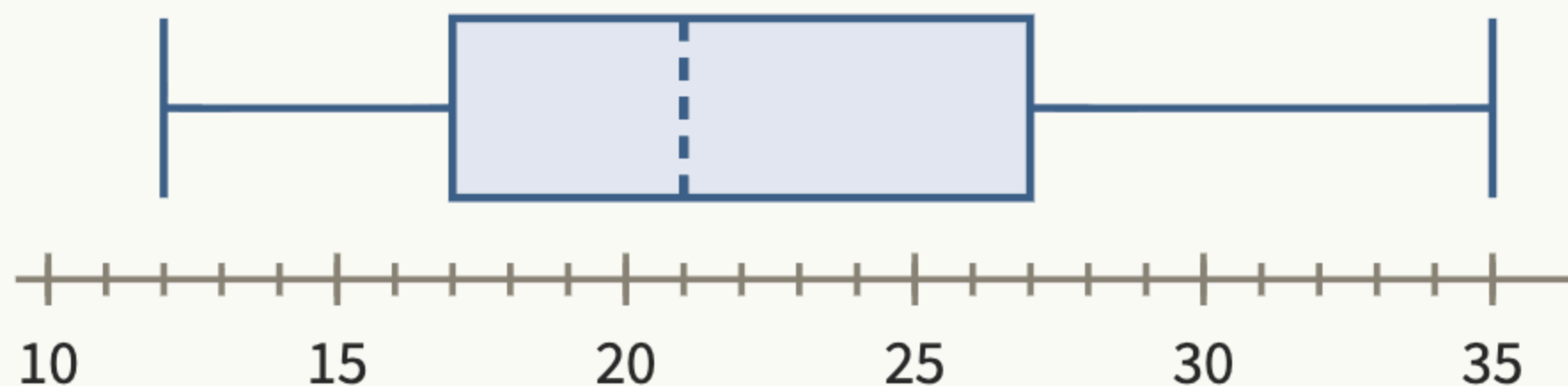
Four Steps to a Boxplot

Procedure 3.4.8 Constructing a Boxplot from a Five-Number Summary

1. Draw a horizontal axis with an appropriate scale that includes both the minimum and the maximum.
2. Draw a box spanning from Q_1 to Q_3 above the axis (the box's width is the IQR).
3. Draw a line inside the box at the median Q_2 .
4. Draw whiskers: short horizontal lines from each end of the box out to the minimum and maximum, with vertical caps marking the endpoints.

From Summary to Picture

The bike shop summary was 12, 17, 21, 27, 35.



The box is 10 units wide. That width is the IQR.

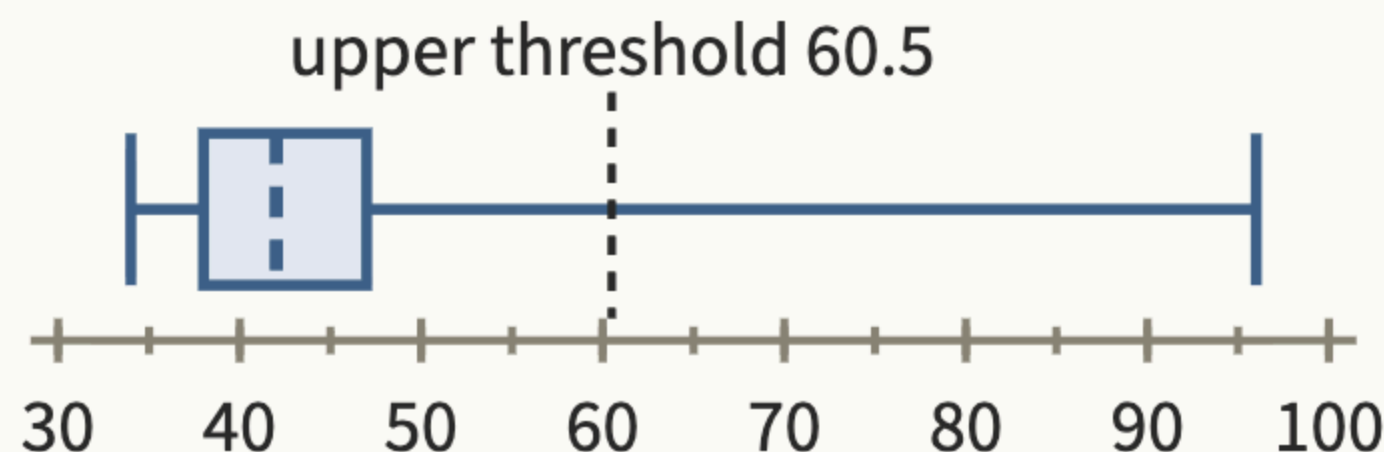
The median is closer to Q_1 than to Q_3 , which suggests the repair counts may be skewed to the right.

What the Picture Tells You

Boxplots provide a quick visual summary of the distribution of data:

- The **box** represents the middle 50% of the data (from Q_1 to Q_3).
- The **line inside the box** represents the median (Q_2).
- The **whiskers** extend from the box to the minimum and maximum values.
- The **width of the box** (the IQR) shows the spread of the middle 50% of the data.
- If the median line is not centered in the box, the data is skewed.

The Whisker Does Not Stop



Warning

In this course the whiskers always run to the minimum and the maximum, so a flagged value gets no special mark. Excel, R, and Python stop the whisker sooner and plot that value as its own dot.

Example 2 flagged 96. The whisker runs straight past 60.5 to reach it.

Now You Try: Summary, Plot, Spread

You Try 3.4.1. Loads run in a dorm laundry room on eleven evenings:

6, 7, 9, 10, 12, 13, 15, 16, 18, 19, 28.

(a) Find the five-number summary.

(b) Sketch the boxplot.

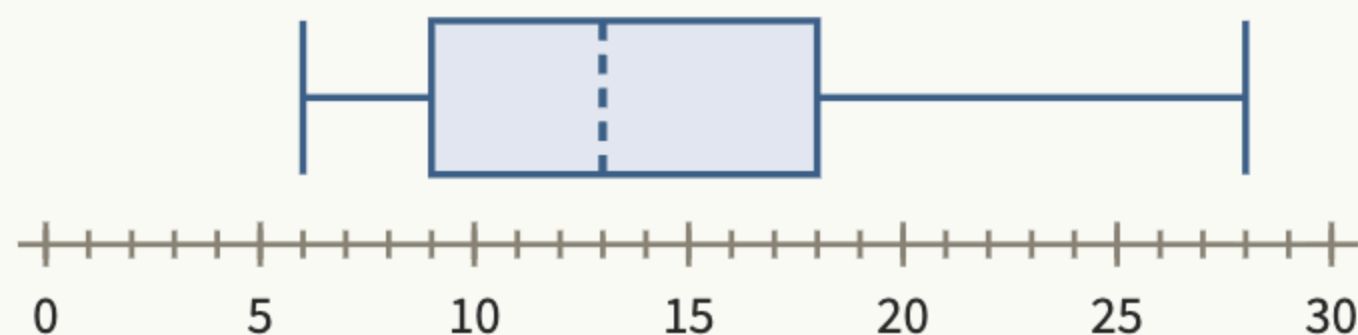
(c) Compute the IQR.

You Try 3.4.1: Checking Your Answers

(a) Five-number summary

There are 11 values, so the median is the 6th: 6, 9, 13, 18, 28

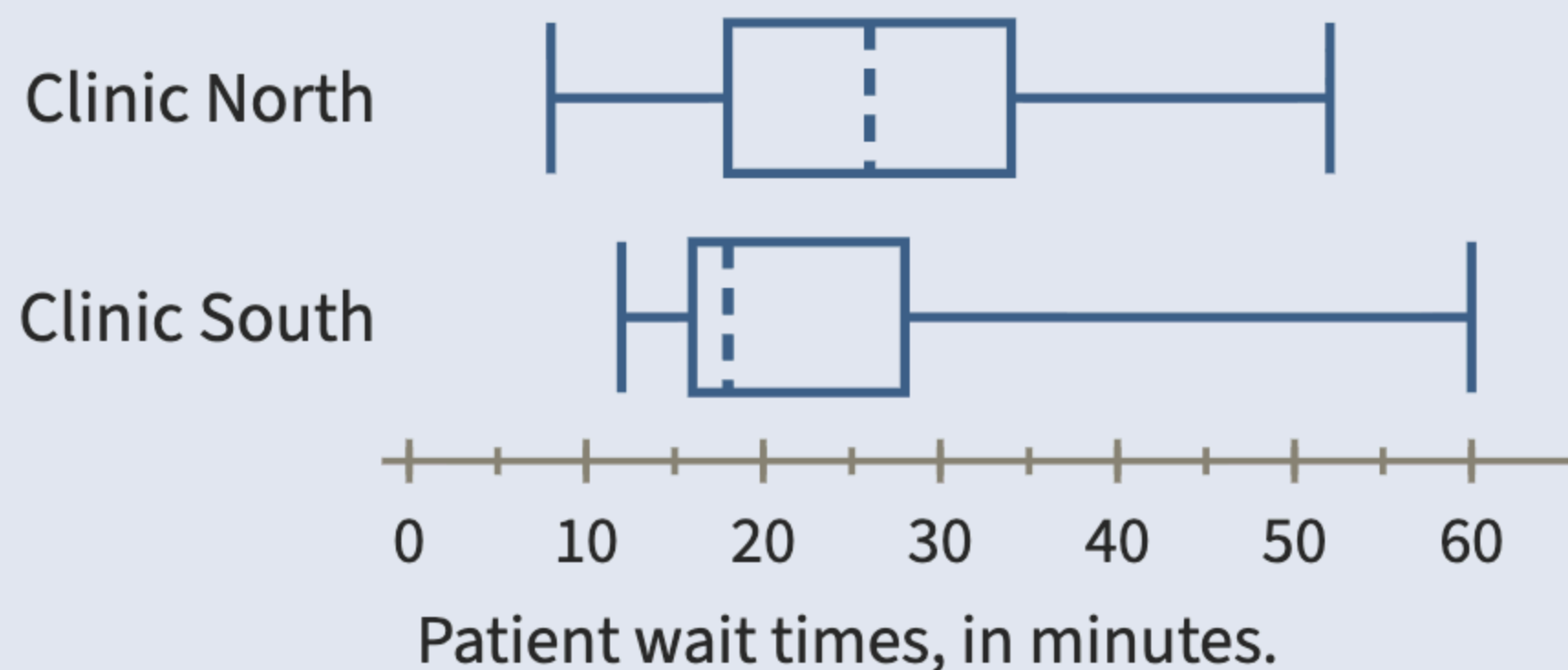
(b) Boxplot



(c) IQR

$$18 - 9 = 9 \text{ loads}$$

Two Clinics, One Decision



**Which line would
you rather stand in?**

Reading Both Summaries

Read the five numbers off each plot.

Clinic	Min	Q_1	Q_2	Q_3	Max
North	8	18	26	34	52
South	12	16	18	28	60

Clinic North: $\text{IQR} = 34 - 18 = 16$ minutes.

Clinic South: $\text{IQR} = 28 - 16 = 12$ minutes.

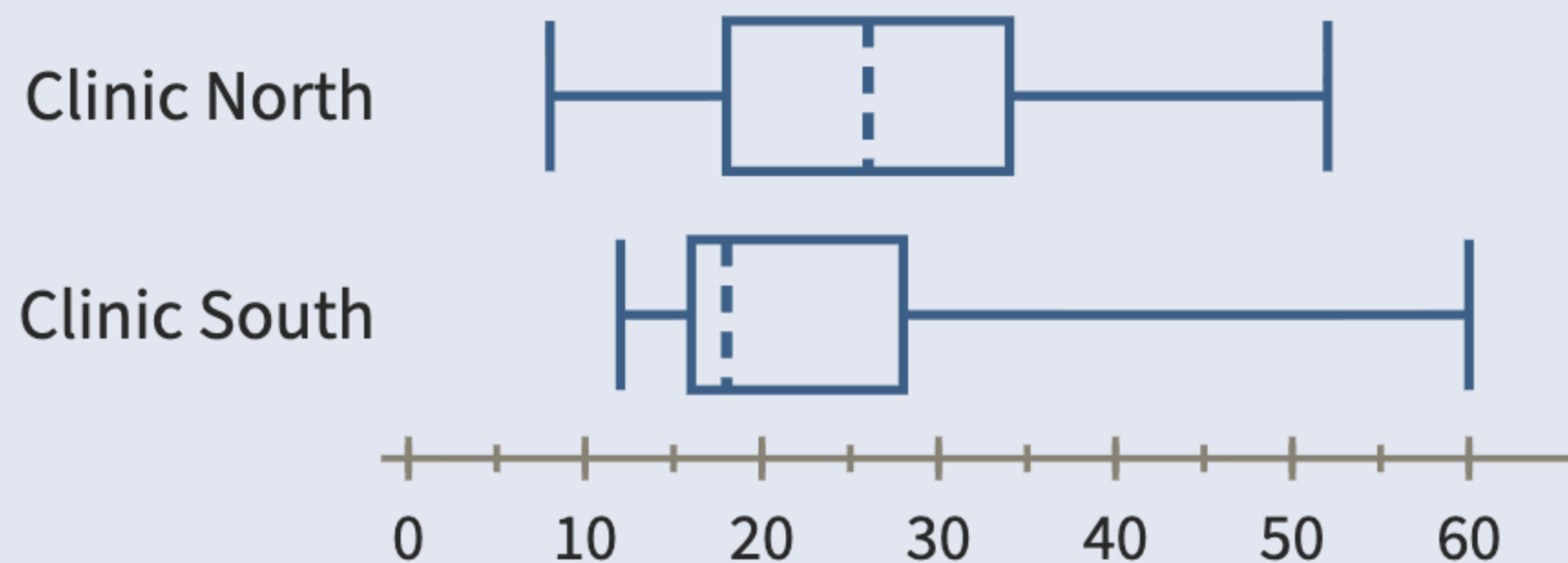
Center, Spread, and Shape

Center: South's median wait is 18 minutes, North's is 26.

Spread: South's middle half is tighter, 12 against 16. South's range is wider, 48 against 44.

Shape: North's median is centered in its box, which suggests North is roughly symmetric. South's is much closer to Q_1 , which suggests South may be skewed to the right.

So Which Line?



South is quicker most days: median 18 against 26, and a tighter middle half.

But South's longest wait is 60 minutes and North's is 52.

Which would you rather risk?

What to Take With You

- Five numbers: Min, Q_1 , Q_2 , Q_3 , Max. Split at the median, and when n is odd, leave the median out of both halves.
- $\text{IQR} = Q_3 - Q_1$ is the width of the middle half. The range is not the same thing.
- The $1.5 \times \text{IQR}$ rule flags a value for a look. The boxplot will not flag it for you.
- Side-by-side boxplots only compare on one shared axis.